

(7) Derivative of Exponential Functions**(1) If $y = e^x$**

$$\text{then } \frac{dy}{dx} = e^x$$

In general* if $y = e^{(f(x))}$

$$\text{then } \frac{dy}{dx} = f'(x) \cdot e^{(f(x))}$$

Example: find $\frac{dy}{dx}$ in the following

$$(1) \quad y = e^{3x}$$

$$\frac{dy}{dx} = (3x)' e^{3x} = 3e^{3x}$$

$$(2) \quad y = e^{\sin(1-x)}$$

$$\frac{dy}{dx} = (\sin(1-x))' e^{\sin(1-x)} = -\cos(1-x) e^{\sin(1-x)}$$

$$(3) \quad y = e^{\sqrt{x}}$$

$$\frac{dy}{dx} = (\sqrt{x})' e^{\sqrt{x}} = \frac{1}{2\sqrt{x}} e^{\sqrt{x}}$$

(2) If $y = a^x$ From **the Result** $a^x = e^{x \ln(a)}$; we can obtain the derivative of $y = a^x$ as follows

$$\begin{aligned} \frac{d}{dx}[a^x] &= \frac{d}{dx}[e^{x \ln(a)}] \\ &= (x \ln(a))' e^{x \ln(a)} \\ &= \ln(a) e^{x \ln(a)} = a^x \ln(a) \end{aligned}$$

In general* If $y = a^{(f(x))}$

$$\text{then } \frac{dy}{dx} = f'(x) \cdot (a^{(f(x))}) \cdot \ln(a)$$

Example: find $\frac{dy}{dx}$ in the following

$$(1) \quad y = a^{3x}$$

$$\frac{dy}{dx} = (3x)' a^{3x} \cdot \ln(a) = 3a^{3x} \ln(a)$$

$$(2) \quad y = 7^{\sin(x)}$$

$$\frac{dy}{dx} = (\sin(x))' 7^{\sin(x)} \cdot \ln(7) = \cos(x) 7^{\sin(x)} \cdot \ln(7)$$

$$(3) \quad y = 5^{\sqrt{x}}$$

$$\frac{dy}{dx} = (\sqrt{x})' 5^{\sqrt{x}} \cdot \ln(5) = \frac{1}{2\sqrt{x}} 5^{\sqrt{x}} \cdot \ln(5)$$

Exercise (7-1)find $\frac{dy}{dx}$ in the following

(1) $y = e^{-3x}$

(2) $y = e^{4x-1}$

(3) $y = e^{-x^2+\sin(x)}$

(4) $y = 3^{(x^2+x)}$

(5) $y = 2^{(x^2-3)}$

(6) $y = 8^{x \sin(x)}$

(8) Derivative of Logarithmic Functions**(1) If $y = \ln(x)$**

$$\text{then } \frac{dy}{dx} = \frac{1}{x}$$

In general* If $y = \ln(f(x))$

$$\text{then } \frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

Example: find $\frac{dy}{dx}$ in the following

$$(1) \quad y = \ln(3x - 1)$$

$$\frac{dy}{dx} = \frac{(3x-1)'}{(3x-1)} = \frac{3}{3x-1}$$

$$(2) \quad y = \ln(\sin(1-x))$$

$$\frac{dy}{dx} = \frac{(\sin(1-x))'}{\sin(1-x)} = -\frac{\cos(1-x)}{\sin(1-x)} = -\cot(1-x)$$

$$(3) \quad y = \ln(e^{1-x})$$

$$\frac{dy}{dx} = \frac{(e^{1-x})'}{(e^{1-x})} = -\frac{e^{1-x}}{(e^{1-x})} = -1$$

$$(4) \quad y = \ln(x^2 + 5x + 2)$$

$$\frac{dy}{dx} = \frac{(x^2+5x+2)'}{(x^2+5x+2)} = \frac{2x+5}{x^2+5x+2}$$

(2) If $y = \log_a(x)$

$$\text{then } \frac{dy}{dx} = \frac{1}{x(\ln(a))}$$

In general* If $y = \log_a(f(x))$

$$\text{then } \frac{dy}{dx} = \frac{f'(x)}{f(x)\ln(a)}$$

Example: find $\frac{dy}{dx}$ in the following

$$(1) \quad y = \log_{10}(15x^2 + 2x)$$

$$\frac{dy}{dx} = \frac{(15x^2+2x)'}{(15x^2+2x)\ln(10)} = \frac{30x+2}{(15x^2+2x)\ln(10)}$$

$$(2) \quad y = \log_5(\sin(x) - e^{-2x})$$

$$\frac{dy}{dx} = \frac{(\sin(x)-e^{-2x})'}{(\sin(x)-e^{-2x})\ln(5)} = \frac{\cos(x)+2e^{-2x}}{(\sin(x)-e^{-2x})\ln(5)}$$

Exercise (7-2)find $\frac{dy}{dx}$ in the following

$$(1) y = \ln(\tan(x)) \quad (2) y = \ln(\sin^2(x)) \quad (3) y = \ln(x^2 + x - 1)$$

$$(4) y = \log_6(\ln(x)) \quad (5) y = \log_3(x + \sqrt{x^2 + 4}) \quad (6) y = \log_2(1 - \cos(x))$$

(9) Derivatives of Hyperbolic Functions

$$(1) \quad y = \sinh(x) = \frac{1}{2}(e^x - e^{-x})$$

$$\frac{dy}{dx} = \frac{1}{2}(e^x + e^{-x}) = \cosh(x)$$

$$(2) \quad y = \cosh(x) = \frac{1}{2}(e^x + e^{-x})$$

$$\frac{dy}{dx} = \frac{1}{2}(e^x - e^{-x}) = \sinh(x)$$

In general

$$* \text{ If } y = \sinh(f(x))$$

$$\text{then } \frac{dy}{dx} = f'(x) \cosh(f(x))$$

$$* \text{ If } y = \cosh(f(x))$$

$$\text{then } \frac{dy}{dx} = f'(x) \sinh(f(x))$$

Note

By using the derivative of $\sinh(x)$ and $\cosh(x)$, we can obtain **the derivative of others trigonometric functions** such as $\tanh(x)$, $\operatorname{sech}(x)$ and $\operatorname{cosech}(x)$

$$(1) \quad y = \tanh(x) = \frac{(e^x - e^{-x})}{(e^x + e^{-x})}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2} \\ &= \frac{e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x}}{(e^x + e^{-x})^2} = \frac{4}{(e^x + e^{-x})^2} = \frac{1}{\left(\frac{1}{2}(e^x + e^{-x})\right)^2} \\ &= \frac{1}{\cosh^2(x)} = \operatorname{sech}^2(x) \end{aligned}$$

In general

$$* \text{ If } y = \tanh(f(x))$$

$$\text{then } \frac{dy}{dx} = f'(x) \cdot \operatorname{sech}^2 h(f(x))$$

Example: find $\frac{dy}{dx}$ in the following

$$(1) \quad y = \operatorname{sech}(x)$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \left[\frac{1}{\cosh(x)} \right] = \frac{\cosh(x)(0) - (1)(\sinh(x))}{\cosh^2(x)} \\ &= \frac{-\sinh(x)}{\cosh^2(x)} = \frac{-1}{\cosh(x)} \frac{\sinh(x)}{\cosh(x)} = -\operatorname{sech}(x) \tanh(x) \end{aligned}$$

$$(2) \quad y = \operatorname{cosech}(x)$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \left[\frac{1}{\sinh(x)} \right] = \frac{\sinh(x)(0) - (1)(\cosh(x))}{\sinh^2(x)} \\ &= \frac{-\cosh(x)}{\sinh^2(x)} = \frac{-1}{\sinh(x)} \frac{\cosh(x)}{\sinh(x)} = -\operatorname{cosech}(x) \coth(x) \end{aligned}$$

$$(3) \quad y = \coth(x)$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \left[\frac{\cosh(x)}{\sinh(x)} \right] = \frac{\sinh(x)(\sinh(x)) - \cosh(\cosh(x))}{\sinh^2(x)} \\ &= 1 - \frac{\cosh^2(x)}{\sinh^2(x)} = 1 - \coth^2(x) \end{aligned}$$

(10) Derivative of Implicit Function

In most of the previous problems, the function is given **explicitly**, i.e. $y = f(x)$, but sometimes the function is given in **an implicit form**, such as

$$(1) \quad x^2 + y^2 = 4xy^3$$

$$(2) \quad xy^2 = 3$$

To find the first derivative of such functions, the first derivative of each term of the equation is taken with respect to the variable x

Example: find $\frac{dy}{dx}$ in the following

$$(1) \quad x^2 + y^2 = y + y^4$$

$$2x + 2y \frac{dy}{dx} = \frac{dy}{dx} + 4y^3 \frac{dy}{dx}$$

$$\frac{dy}{dx} (2y - 1 - 4y^3) = -2x$$

$$\therefore \frac{dy}{dx} = \frac{-2x}{2y-1-4y^3}$$

$$(2) \quad x = \cos(xy)$$

$$1 = -\left(x \frac{dy}{dx} + y\right) \sin(xy)$$

$$x \frac{dy}{dx} \sin(xy) = -1 - y \sin(xy)$$

$$\therefore \frac{dy}{dx} = \frac{-1-y \sin(xy)}{x \sin(xy)}$$

Exercise (7-3)

find $\frac{dy}{dx}$ in the following

$$(1) \quad \sqrt{x} + \sqrt{y} = 2$$

$$(2) \quad x^{-1} + y^{-1} = 1$$

$$(3) \quad \cos(x^2 + y^2) = 2x$$

$$(4) \quad (x + y)^2 = x^2 + y^2$$

$$(5) \quad xy - x^3 + y^2 = 0$$

$$(6) \quad x = y\sqrt{1 - y^2}$$